

Project Tracking Platform — Full Planning Document

Author: Milton | **Status:** Draft | **Version:** 1.0

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1. Problem & Goal

Problem

The company has no centralized way to track:

- Project status and Technology Readiness Level (TRL 1–9)
- Hours invested per project and per person
- Who is working on what

- Milestones and deadlines
- Task-level progress (Kanban)

Goal

Build an internal web app that gives employees, managers, and admins a **single source of truth** for all project activity with hour logging, TRL tracking, milestone management, a Kanban board, and exportable reports.

2. Full Feature List



Authentication

- ☐ User registration (name, email, password, role)
- ☐ Secure login with JWT tokens
- ☐ Password hashing (bcrypt)
- ☐ Role-based access control (Employee / Manager / Admin)
- ☐ Logout + token expiry handling
- ☐ Admin can deactivate/reactivate accounts



Project Management

- ☐ Create, edit, archive projects
- ☐ Project fields: name, description, start date, status, TRL level, team members
- ☐ Search and filter projects by status, TRL, team
- ☐ Assign / unassign team members to projects
- ☐ Project detail view with overview of all related data



Hour Logging

- ☐ Log entry fields: project, date, hours, optional notes
- ☐ Log hours per day per project
- ☐ Edit and delete own log entries

- ☐ View personal hours (daily / weekly / monthly)
- ☐ Managers see all hours for their projects
- ☐ Admins see all hours across all projects

Dashboard

- ☐ Overview of all active projects
- ☐ Each project card shows: TRL level, total hours, team size, status
- ☐ Quick stats: total hours this week, active projects, upcoming milestones
- ☐ Overdue milestones flagged visually
- ☐ Recent activity feed

TRL Tracker

- ☐ Display current TRL level (1–9) per project
- ☐ Update TRL with timestamp + justification note
- ☐ Full TRL history log per project
- ☐ Visual progress indicator (e.g. progress bar or stage badge)

Milestone Tracking

- ☐ Add milestones to a project (name, due date, status)
- ☐ Milestone statuses: Pending / In Progress / Completed
- ☐ Visual flag for overdue milestones
- ☐ Milestone history retained after completion

Kanban Board (Monday.com-style)

- ☐ Per-project Kanban board with columns (e.g. To Do / In Progress / Review / Done)
- ☐ Cards represent tasks with: title, description, assignee, due date, priority
- ☐ Drag-and-drop cards between columns
- ☐ Add, edit, delete cards

- ☐ Filter cards by assignee or priority
- ☐ Custom column creation per project
- ☐ Optional: Global Kanban view across all projects

Reports

- ☐ Hours per person (total, weekly, monthly)
- ☐ Hours per project breakdown
- ☐ Project status overview report
- ☐ TRL progression report per project
- ☐ Export reports to PDF
- ☐ Export reports to Excel (.xlsx)

Team & People

- ☐ View all users in the organization
- ☐ View a person's assigned projects and logged hours
- ☐ Admin can manage user roles and accounts

Admin Panel

- ☐ User management (create, edit, deactivate users)
- ☐ Assign roles
- ☐ View system-wide data
- ☐ Manage all projects (full access)

3. MVP vs Future Releases

MVP (Version 1.0) — Build This First

The MVP focuses on the core loop: log in → see projects → log hours → track TRL.

Feature	Priority
Authentication (login, register, roles)	🔴 Must Have
Project creation & management	🔴 Must Have
Hour logging (per day per project)	🔴 Must Have
Main dashboard (project overview)	🔴 Must Have
TRL level tracking + history	🔴 Must Have
Milestone tracking (basic)	🔴 Must Have
Hours report by person (view on screen)	🔴 Must Have
Admin user management	🔴 Must Have



Version 1.5 — First Enhancement

Feature	Priority
Kanban board per project	🟡 High
PDF + Excel export	🟡 High
Advanced filters on dashboard	🟡 Medium
Personal profile page	🟡 Medium



Version 2.0 — Future

Feature	Priority
Global Kanban across all projects	🟡 Medium
Email notifications for overdue milestones	🟡 Medium
Activity feed / audit log	🟡 Medium
Mobile responsive improvements	🟡 Medium
Budget / cost tracking	🔵 Nice to Have
Slack / Google Workspace integration	🔵 Nice to Have
Dark mode	🔵 Nice to Have

4. All Pages & Screens

Public (No Login Required)

Page	Route	Description
Login	<code>/login</code>	Email + password login form
Register	<code>/register</code>	New user sign-up (or admin-only invite)
Forgot Password	<code>/forgot-password</code>	Password reset flow

Employee Pages

Page	Route	Description
Dashboard	<code>/dashboard</code>	Overview of all projects + personal stats
My Hours	<code>/my-hours</code>	Personal hour log — view, add, edit entries
Log Hours	<code>/log-hours</code>	Quick form to log hours for today
Projects	<code>/projects</code>	List of all projects (searchable, filterable)
Project Detail	<code>/projects/:id</code>	Full project info, TRL, milestones, team
Kanban Board	<code>/projects/:id/kanban</code>	Per-project task board (Monday.com-style)
My Profile	<code>/profile</code>	Personal info, hours summary, assigned projects

Project Manager Pages

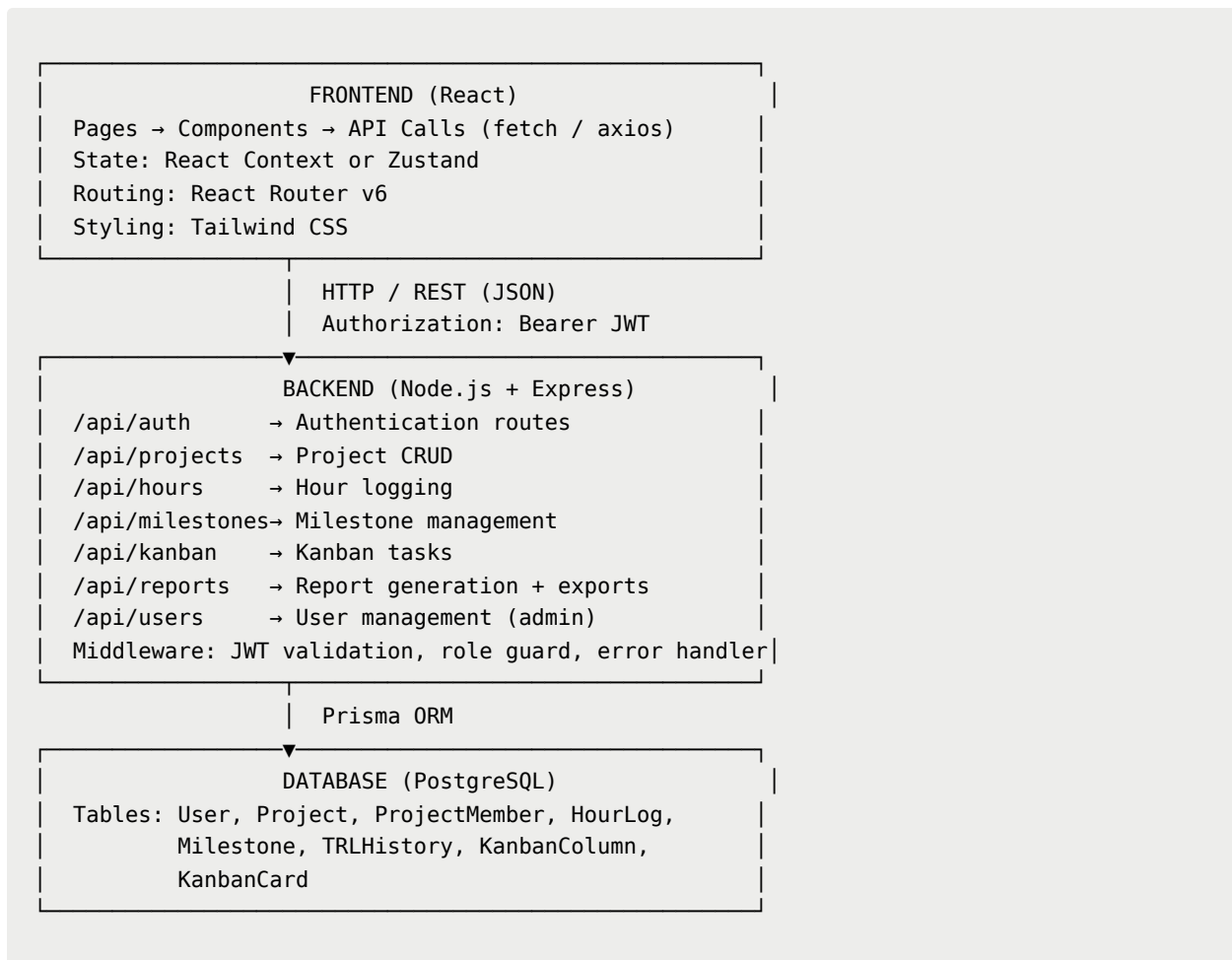
Page	Route	Description
All of Employee pages	—	Full access to employee views
Manage Project	<code>/projects/:id/edit</code>	Edit project details, TRL, milestones
Project Hours	<code>/projects/:id/hours</code>	View all hours logged by team on this project
Team	<code>/projects/:id/team</code>	Assign/unassign team members
Milestones	<code>/projects/:id/milestones</code>	Add and manage milestones
Reports	<code>/reports</code>	View hours by person/project, export PDF/Excel

Admin Pages

Page	Route	Description
All of Manager pages	—	Full access to all views
Admin Panel	<code>/admin</code>	System overview, quick actions
User Management	<code>/admin/users</code>	Create, edit, deactivate users, assign roles
All Projects	<code>/admin/projects</code>	Full access to all projects
All Reports	<code>/admin/reports</code>	Company-wide hours + project reports
Settings	<code>/admin/settings</code>	App configuration (TRL labels, column names, etc.)

Total Pages: ~18 unique screens

5. System Architecture



Folder Structure (Suggested)

```
/client          ← React frontend
  /src
    /pages        ← One file per page/route
    /components   ← Reusable UI components
    /hooks        ← Custom React hooks
    /context      ← Auth context, global state
    /services     ← API call functions
    /utils        ← Helpers, formatters

/server          ← Express backend
  /routes         ← Route handlers per resource
  /controllers    ← Business logic
  /middleware     ← Auth guard, role guard
  /prisma         ← Schema + migrations
  /services       ← Report generation, exports
  /utils         ← Helpers
```

6. Data Models

User

```
id, name, email, passwordHash, role (EMPLOYEE | MANAGER | ADMIN),
createdAt, isActive
```

Project

```
id, name, description, startDate, status (ACTIVE | ARCHIVED | COMPLETED),
currentTRL (1-9), createdBy (userId), createdAt, updatedAt
```

ProjectMember

```
id, projectId, userId, joinedAt
```

HourLog


```
id, userId, projectId, date, hours, notes, createdAt
```

Milestone

```
id, projectId, name, dueDate, status (PENDING | IN_PROGRESS | COMPLETED),  
completedAt, notes
```

TRLHistory

```
id, projectId, trlLevel, updatedBy (userId), justification, recordedAt
```

KanbanColumn

```
id, projectId, title, order
```

KanbanCard

```
id, columnId, projectId, title, description, assigneeId,  
dueDate, priority (LOW | MEDIUM | HIGH), order, createdAt
```

7. API Endpoints Overview

Auth

```
POST /api/auth/register  
POST /api/auth/login  
POST /api/auth/logout  
GET /api/auth/me
```

Projects

```
GET    /api/projects          ← list all (with filters)
POST   /api/projects          ← create
GET     /api/projects/:id     ← get one
PUT     /api/projects/:id     ← update
DELETE  /api/projects/:id     ← archive

GET     /api/projects/:id/members ← get team
POST    /api/projects/:id/members ← assign member
DELETE  /api/projects/:id/members/:userId ← remove
```

Hours

```
GET     /api/hours          ← my hours (query: project, date range)
POST    /api/hours          ← log entry
PUT     /api/hours/:id      ← edit entry
DELETE  /api/hours/:id      ← delete entry
GET     /api/hours/project/:id ← all hours for a project (manager/admin)
GET     /api/hours/user/:id  ← all hours for a user (admin)
```

Milestones

```
GET     /api/projects/:id/milestones
POST    /api/projects/:id/milestones
PUT     /api/milestones/:id
DELETE  /api/milestones/:id
```

TRL

```
GET     /api/projects/:id/trl ← full TRL history
POST    /api/projects/:id/trl ← update TRL level
```

Kanban

```
GET     /api/projects/:id/kanban ← columns + cards
POST    /api/projects/:id/kanban/columns ← add column
PUT     /api/kanban/columns/:id ← rename/reorder
DELETE  /api/kanban/columns/:id
POST    /api/projects/:id/kanban/cards ← add card
```

```
PUT    /api/kanban/cards/:id          ← edit / move card
DELETE /api/kanban/cards/:id
```

Reports

```
GET    /api/reports/hours-by-person   ← query: range
GET    /api/reports/hours-by-project
GET    /api/reports/project-status
GET    /api/reports/export/pdf
GET    /api/reports/export/excel
```

Admin / Users

```
GET    /api/users
POST   /api/users
PUT    /api/users/:id
PATCH /api/users/:id/deactivate
```

8. Kanban Module (Monday.com-style)

How it works

Each project has its own Kanban board with customizable columns. A card represents a task and can be dragged between columns to update its status.

Default Columns (customizable)

1. **To Do** — tasks not yet started
2. **In Progress** — currently being worked on
3. **In Review** — waiting for feedback or approval
4. **Done** — completed tasks

Card Fields

- Title (required)
- Description

- Assignee (team member on this project)
- Due date
- Priority: Low / Medium / High (color-coded)
- Column (status)

Key Interactions

- Drag and drop cards between columns (use `@dnd-kit/core` library)
- Click card to open detail modal
- Filter by assignee or priority
- Add/rename/delete columns
- Order cards within a column by priority or due date

Suggested Library

Use `@dnd-kit/core` for drag-and-drop (lightweight, works well with React, no jQuery dependency).

9. User Roles & Permissions Matrix

Action	Employee	Manager	Admin
View all projects	✓	✓	✓
Create project	✗	✓	✓
Edit project	✗	✓ (own)	✓
Archive project	✗	✓ (own)	✓
Log own hours	✓	✓	✓
Edit own hours	✓	✓	✓
View team hours	✗	✓ (own project)	✓
Update TRL level	✗	✓ (own project)	✓
Manage milestones	✗	✓ (own project)	✓
View Kanban	✓	✓	✓
Manage Kanban cards	✓	✓	✓

Action	Employee	Manager	Admin
Manage Kanban columns	✗	✓	✓
Export reports	✗	✓ (own project)	✓
Manage users	✗	✗	✓
Access admin panel	✗	✗	✓

10. Tech Stack Summary

Layer	Choice	Why
Frontend	React.js + Tailwind CSS	Fast dev, component-based, great ecosystem
Routing	React Router v6	Standard for React SPAs
State	React Context + Zustand	Simple global state without Redux overhead
Backend	Node.js + Express.js	JS full-stack, lightweight, fast to build APIs
ORM	Prisma	Type-safe, easy migrations, readable schema
Database	PostgreSQL	Relational, reliable, perfect for this data model
Auth	JWT (jsonwebtoken)	Stateless, easy RBAC, widely supported
Drag & Drop	@dnd-kit/core	Lightweight D&D for Kanban
PDF Export	Puppeteer or pdfkit	Server-side PDF generation
Excel Export	ExcelJS	.xlsx generation in Node.js
Hosting	Railway or Render	Simple deploy, free tier, supports Node + Postgres

11. Open Questions

#	Question	Owner	Status
1	What are the internal TRL level definitions used by the company?	Milton + Supervisor	🔴 Open
2	Should hour logs require manager approval before being final?	Stakeholder	🔴 Open

#	Question	Owner	Status
3	Is the Kanban board scoped per project only, or also a global view?	Employee / Stakeholder	● TBD
4	What is the expected number of users and projects?	IT / Admin	● TBD
5	Where will the app be hosted (cloud, internal server)?	IT	● Open
6	Should non-assigned employees be able to view all projects, or only their own?	Stakeholder	● TBD
7	Will Kanban cards need file attachments or comments?	Employee feedback	● Future
8	Should there be email notifications for overdue milestones?	Stakeholder	● Future

Last updated: June 2026 — Milton

UI design